

Optimisation of Cosolvent Concentration for Topical Drug Delivery – II: Influence of Propylene Glycol on Ibuprofen Permeation

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Key Words

Ibuprofen · Propylene glycol · Silicone · Skin · Vehicle composition

Abstract

The influence of formulation components on drug thermodynamic activity and on membrane properties has not been extensively studied to date. Propylene glycol (PG) has a long history of use as a component of topical pharmaceutical and cosmetic preparations. In the present study we investigated the influence of PG on the solubility and membrane permeability characteristics of ibuprofen in silicone and skin using binary (PG:water) and ternary (ethanol:PG:water) solvent systems. Fluxes were maximum for 70:30 PG:water systems in silicone membrane; however, for experiments conducted with skin, the flux of ibuprofen systematically increased with increasing amounts of PG. For silicone membrane, the flux values of ibuprofen from ternary ethanol:PG:water (25:25:50 and 50:25:25) systems were 1.2- and 1.5-fold higher than the highest values observed from the PG:water systems. A comparison of the data from occluded versus non-occluded studies suggested that the ethanol content was the critical determinant of flux enhancement in these systems. For human skin, a PG:water (50:50) mixture enhanced ibuprofen flux to the same extent as the corresponding ethanol:PG:water (25:25:50) system. However, the ternary system, eth-

anol:PG:water (50:25:25), demonstrated the greatest enhancement. An analysis of the partition and diffusion parameters for experiments conducted with skin confirmed that the influence of PG in the PG:water systems is primarily on the solubility and partitioning behaviour of ibuprofen. The ternary systems also appear to influence skin partition behaviour rather than diffusivity when compared with the permeation parameters of ibuprofen from saturated aqueous solutions.

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Introduction

Propylene glycol (PG) is a colourless, viscous liquid which was described originally by Wurtz [1] in 1859. Its first recorded application in pharmaceutical preparations was in 1932 and, currently, PG is widely used as a cosolvent for poorly soluble materials and/or to enhance drug permeation through skin from topical preparations [2–6]. Despite its long history of use, the precise role of PG in enhancing drug permeation remains to be clarified. The literature details conflicting reports concerning its effects on human and animal skin. Aguiar and Weiner [7] studied the effect of PG on chloramphenicol permeation in hairless mouse skin and concluded that PG did not alter membrane properties. From studies on fluo-

cinolone acetonide permeability in human skin, Ostren-ga et al. [8] observed that pre-treatment of skin with pure PG caused the skin barrier to become less pliable and attributed this effect to dehydration. Zatz and Dalvi [9] investigated benzocaine permeation in hairless mouse skin from a range of solvents and showed that, with water as a standard, interaction effects were not significant when PG was used as a solvent. Hoelgaard and Mollgaard [10] examined the effects of PG on metronidazole flux enhancement in human skin and suggested a possible 'carrier-solvent' effect. Kasting et al. [11] demonstrated significant permeation enhancement of triprolidine base from pure PG when compared with oil vehicles in human skin. These workers related this direct effect of PG on the membrane to solvent incorporation in the stratum corneum in a sufficient quantity to modify permeant solubility in the tissue.

As demonstrated previously, topical presentations of agents like ibuprofen which are intended for subcutaneous action cannot be assumed to be pharmaceutically and clinically equivalent or indeed interchangeable [12]. Moreover, although it is evident that different penetration enhancers influence drug diffusion characteristics to different extents [13], the complexity of most pharmaceutical preparations makes it difficult to deconvolve the exact influence of specific enhancers on drug permeation. Recently we reported the results of a systematic investigation of the influence of binary systems of ethanol and water on ibuprofen solubility and in vitro permeation characteristics in model membranes and human skin [14]. Analysis of the data confirmed that increasing ethanol content enhanced the solubility of ibuprofen in both membranes as well as in the vehicle itself. The aim of this work was to extend these studies to explore the influence of PG on ibuprofen permeation and to investigate whether specific effects on partition and diffusion parameters may be distinguished. A secondary goal of the study was to investigate the effects of ethanol in more complex ternary systems (PG:ethanol:water) on permeation in silicone and skin, as such mixtures are more representative of commercial formulations than simple binary systems.

Materials and Methods

Materials

Ibuprofen was a generous gift from Wyeth Consumer Health Care (Havant, UK). HPLC grade solvents were obtained from Fisher Scientific, UK. All other materials were of reagent grade and were obtained from VWR, UK.

Solubility Determination

Excess drug was added to each solvent or cosolvent mixture and stirred with a magnetic bar for 48 h in a water bath maintained at 32°C. Solutions were centrifuged for 15 min at 10,000 rpm. The supernatant was then diluted and assayed either by HPLC or UV spectroscopy. For HPLC analysis, the sample was diluted in isopropanol, followed by appropriate dilution in methanol. The stationary phase was an Apex reverse-phase ODS 5- μ m packed column (250 $\times$ 4.6 mm) and a mobile phase of 65% acetonitrile and 35% phosphate buffer (adjusted to pH 3.2 using orthophosphoric acid). UV detection was performed at 225 nm and a flow rate of 1.2 ml min⁻¹. The retention time was approximately 5 min, and calibration curves were constructed on the basis of peak area measurements using >5 standard solutions. Experiments were performed in triplicate and mean values with standard deviation and coefficient of variation were calculated. Reproducibility was evaluated prior to injection of the samples and during analysis; coefficients of variation were <5% in all experiments.

Permeation Studies in Silicone Membrane and Skin

The silicone membranes used in this work were obtained from Samco (Nuneaton, UK) with a measured thickness of 400 μ m. Full-thickness human skin from cosmetic surgery was obtained by informed consent and with appropriate institutional ethical approval. The epidermis was separated by immersing the skin in water maintained at 60°C for 1 min [15]. It was then pinned to a corkboard and the epidermis was carefully peeled away from the dermis. The epidermis was then immersed in cold water to aid mounting onto filter paper and stored in a freezer until required. Prior to the diffusion experiment the skin was defrosted and cut to appropriate size.

Permeation experiments were conducted using saturated solutions of ibuprofen for each vehicle using Franz diffusion cells with a receptor volume of 2.5 ml, a diffusion area of 1 cm² and an application volume of 1 ml. For all permeation experiments the receptor fluid was pH 7.4 phosphate-buffered saline and sink conditions were maintained at all time points. The diffusion experiments were performed under occluded conditions by sealing the donor compartment with microscope coverslips. The ternary systems were studied under both occluded and non-occluded conditions. The number of replicates was at least 4.

Fluxes were calculated by monitoring the cumulative amount (Q) of drug diffused as a function of time (t). The permeability coefficient (k_p) was derived from the linear portion of this profile when steady-state transport had been achieved. Partition ($\alpha = Kh$) and diffusion parameters ($\beta = D/h^2$) were derived by fitting the entire Q versus t profile to equation 1 [16].

$$Q = c_v \alpha \left[\beta t - \frac{1}{6} - \frac{2}{\pi^2} \sum_{n=1}^{\infty} \frac{(-1)^n}{h^2} \exp(n^2 \pi^2 \beta t) \right] \quad (1)$$

where K is the membrane-vehicle partition coefficient of the drug, D is its diffusivity in the membrane of thickness h, α represents Kh, and β represents D/h^2 . Differences between parameters were evaluated using ANOVA with the level of significance set at $p \leq 0.05$.

Results

Solubility Studies

Figure 1 shows the exponential increase in ibuprofen solubility with increasing PG content. Above 60% v/v PG there is a marked increase in the solubility of ibuprofen; the drug's solubility in pure PG is $\sim 1,750$ times greater than its aqueous solubility (fig. 1). This exponential increase has also been reported by Manrique et al. [17], who investigated the solubility of ibuprofen at several temperatures in PG-water cosolvent mixtures. The solubilities of ibuprofen in the ethanol:PG:water systems studied are given in table 1. These values are approximately 1,500-fold and 3,000-fold greater, for the 25:25:50 system and the 50:25:25 systems, respectively, than the aqueous solubility of ibuprofen (0.09 mg/ml) [14].

Permeation of Ibuprofen from Binary and Ternary Systems in Silicone

The permeation of ibuprofen through silicone membrane from PG:water vehicles of varying composition was evaluated. Table 2 lists the fluxes, kinetic parameters (D/h^2), thermodynamic parameters ($K \times h$) and the permeability coefficients obtained (k_p). The steady-state fluxes varied from $\sim 300 \mu\text{g}/\text{cm}^2/\text{h}$ in pure PG to $185 \mu\text{g}/\text{cm}^2/\text{h}$ in water. At 60–70% PG (v/v) the flux was maximal and approximately double that of the drug from water. There are no clear trends for the kinetic and thermodynamic parameters, but permeability coefficients appear to decrease above 60% PG.

The permeation of ibuprofen through silicone membrane, using ternary vehicles composed of ethanol, PG and water was further evaluated. Table 3 lists the kinetic parameters (D/h^2), thermodynamic parameters (Kh) and permeability coefficients (k_p). The fluxes of ibuprofen from ternary formulations are not significantly different. There are significant differences ($p < 0.05$) in the thermodynamic and kinetic parameters for the two formulations studied and the permeability coefficient was significantly lower for the formulation containing the greater amount of ethanol.

To simulate better how such formulations are used on skin, diffusion studies for the ternary vehicles were also conducted on silicone membrane under non-occluded conditions. As for the occluded studies, 1 ml of a saturated solution of ibuprofen in the relevant formulation was applied. While the fluxes for these formulations were the same under occluded conditions, there was a significant increase ($p < 0.05$) in flux for the formulation containing more ethanol under non-occluded conditions when com-

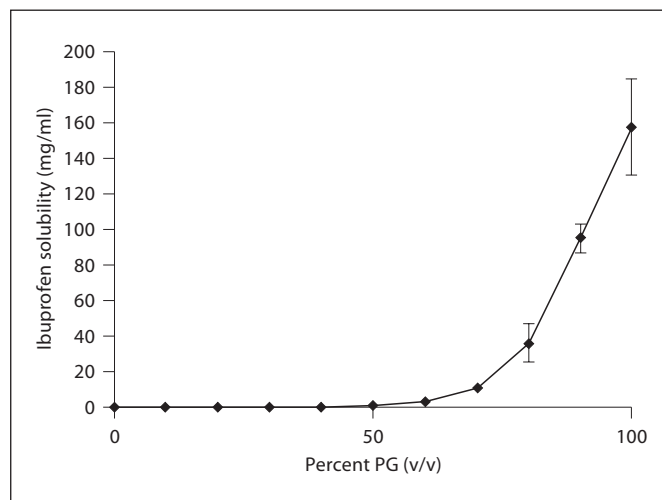


Fig. 1. Solubility of ibuprofen in PG and water mixtures at 32°C ($n = 3$).

Table 1. Solubility of ibuprofen in ethanol/PG/water mixtures

Ethanol:PG:water	Solubility, mg/ml
25:25:50	133 $\pm$ 25
50:25:25	292 $\pm$ 5

pared with the system with lower ethanol. In addition, the flux from the system with lower ethanol content is significantly greater for occluded versus non-occluded conditions ($p < 0.05$). The k_p value for the non-occluded system with lower ethanol is significantly lower than the corresponding k_p value for the occluded system (tables 3, 4). No differences are seen in the k_p values for the systems with higher ethanol content and fluxes for the formulations with the higher proportion of ethanol are statistically the same whether occluded or non-occluded.

Permeation of Ibuprofen from Binary and Ternary Systems in Skin

As expected, the fluxes for ibuprofen permeating through human skin were considerably lower than for the corresponding formulations through silicone membrane (tables 5, 6). For the binary PG:water systems, as the amount of PG increased, so did the permeation rate. The flux increase is particularly noticeable between 25 and 50% PG. There was no significant difference in the ibuprofen fluxes from the PG:water 50:50 mixture and from

Table 2. Steady-state fluxes, thermodynamic and kinetic parameters and permeability coefficients of ibuprofen through silicone membrane from saturated formulations containing PG and water (mean $\pm$ SD, n = 4)

PG:water (v/v)	J, $\mu\text{g}/\text{cm}^2/\text{h}$	$K \times h$, μm	D/h^2 , h^{-1}	k_p , cm/h
0:100	185.5 $\pm$ 10.94	1.83 $\pm$ 0.35	1,368 $\pm$ 684	2.09 $\pm$ 0.14
10:90	225.4 $\pm$ 31.47	2.16 $\pm$ 1.39	2,052 $\pm$ 828	3.55 $\pm$ 0.86
20:80	229.5 $\pm$ 14.20	1.03 $\pm$ 0.44	821 $\pm$ 115	2.09 $\pm$ 0.77
30:70	216.0 $\pm$ 22.47	1.01 $\pm$ 0.06	1,278 $\pm$ 171	1.35 $\pm$ 0.13
40:60	277.1 $\pm$ 31.22	0.76 $\pm$ 0.35	882 $\pm$ 260	0.59 $\pm$ 0.05
60:40	355.4 $\pm$ 36.34	0.05 $\pm$ 0.01	1,933.2 $\pm$ 543.6	0.11 $\pm$ 0.01
70:30	443.3 $\pm$ 19.86	15.11 $\pm$ 2.04	37.4 $\pm$ 7.92	0.04 $\pm$ 0.01
80:20	245.2 $\pm$ 28.55	$1.38 \times 10^{-2} \pm 7.21 \times 10^{-3}$	504 $\pm$ 288	$6.70 \times 10^{-3} \pm 7.31 \times 10^{-4}$
90:10	219.6 $\pm$ 22.32	0.37 $\pm$ 0.05	6.30 $\pm$ 4.32	$2.31 \times 10^{-3} \pm 2.35 \times 10^{-4}$
100:0	297.1 $\pm$ 29.43	1.85 $\pm$ 1.1	1.02 $\pm$ 0.36	$1.88 \times 10^{-3} \pm 1.81 \times 10^{-4}$

Table 3. Steady-state fluxes, thermodynamic and kinetic parameters and permeability coefficients of ibuprofen through silicone membrane from saturated formulations containing ethanol, PG and water (mean $\pm$ SD, n = 4)

EtOH:PG:water (v/v)	J, $\mu\text{g}/\text{cm}^2/\text{h}$	$K \times h$, μm	D/h^2 , h^{-1}	k_p , cm/h
25:25:50	525.8 $\pm$ 37.3	$8.9 \times 10^{-3} \pm 1.5 \times 10^{-3}$	421.2 $\pm$ 0.7	$3.7 \times 10^{-3} \pm 3.3 \times 10^{-8}$
50:25:25	641.1 $\pm$ 84.1	2.4 $\pm$ 0.2*	0.93 $\pm$ 0.19*	$2.2 \times 10^{-3} \pm 8.7 \times 10^{-5*}$

* $p < 0.05$.

Table 4. Steady-state fluxes, thermodynamic and kinetic parameters and permeability coefficients of ibuprofen through silicone membrane from saturated formulations containing ethanol, PG and water under non-occluded conditions (mean $\pm$ SD, n = 4)

EtOH:PG:water (v/v)	J, $\mu\text{g}/\text{cm}^2/\text{h}$	$K \times h$, μm	D/h^2 , h^{-1}	k_p , cm/h
25:25:50	376.2 $\pm$ 39.98	4.2 $\pm$ 0.8	0.7 $\pm$ 0.2	$2.9 \times 10^{-3} \pm 2.8 \times 10^{-4}$
50:25:25	558.2 $\pm$ 48.94*	1.6 $\pm$ 1.2*	1.1 $\pm$ 0.1*	$1.9 \times 10^{-3} \pm 1.2 \times 10^{-4*}$

* $p < 0.05$.

Table 5. Steady-state fluxes, thermodynamic and kinetic parameters and permeability coefficients of ibuprofen through human epidermis from saturated formulations containing PG and water (mean $\pm$ SD, n = 5)

PG:water (v/v)	J, $\mu\text{g}/\text{cm}^2/\text{h}$	$K \times h$, μm	D/h^2 , h^{-1}	k_p , cm/h
0:100	24.4 $\pm$ 2.69	1,183 $\pm$ 503	0.23 $\pm$ 0.02	0.27 $\pm$ 0.03
25:75	35.5 $\pm$ 9.77	460 $\pm$ 260	0.7 $\pm$ 0.3**	0.26 $\pm$ 0.08
50:50	99.1 $\pm$ 22.77*	2.5 $\pm$ 0.2*	0.4 $\pm$ 0.1**	$9.58 \times 10^{-4} \pm 4.86 \times 10^{-5***}$
100:0	133.8 $\pm$ 36.1*	1.36 $\pm$ 0.15*	0.6 $\pm$ 0.4**	$8.24 \times 10^{-4} \pm 1.75 \times 10^{-4***}$

* $p < 0.05$ (comparison to 0:100 and 25:75 systems); ** $p < 0.05$ (comparison to 0:100 system); *** $p < 0.05$ (comparison to 0:100 and 25:75 systems).

Table 6. Steady-state fluxes, thermodynamic and kinetic parameters and permeability coefficients of ibuprofen through human epidermis from saturated formulations containing ethanol, PG and water through human epidermis (mean $\pm$ SD, n = 5)

EtOH:PG:water (v/v)	J, $\mu\text{g}/\text{cm}^2/\text{h}$	$K \times h$, μm	D/h^2 , h^{-1}	k_p , cm/h
25:25:50	77.6 ± 9.04	1.07 ± 0.94	0.63 ± 0.25	$6.70 \times 10^{-4} \pm 9.82 \times 10^{-5}$
50:25:25	$203.5 \pm 36.24^*$	$2.83 \pm 0.30^*$	$0.25 \pm 0.02^*$	$7.09 \times 10^{-4} \pm 1.24 \times 10^{-4}$

* $p < 0.05$.

pure PG. The thermodynamic parameters derived for the formulations containing 0 and 25% PG are not significantly different, but there is a significant difference in these parameters compared with the 50% PG mixture and pure PG. The kinetic parameter does not alter significantly across the range of formulations used in the study. The results of permeation studies conducted using ternary solvent formulations applied to human skin are given in table 6. In contrast to the silicone data, ibuprofen flux was almost threefold higher from the ethanol-rich formulation relative to that from the vehicle with less of the volatile solvent. The thermodynamic and kinetic parameters are significantly different for the two vehicles ($p < 0.05$).

Discussion

PG is a common component of many cosmetic and pharmaceutical formulations, but the mechanism underlying its ability to modulate active permeation has not been explored in detail to date. A better understanding of precisely how PG influences active permeation should assist in the design of more effective vehicles for delivery to and through the skin. In this study, the influence of PG on ibuprofen solubility and the permeation of ibuprofen from binary and ternary solvent systems in silicone membrane and in skin were explored in order to gain insight into PG enhancement properties.

The solubility of ibuprofen increased exponentially with PG in line with the results of Herkenne et al. [18]. Solute-solvent interactions are greater in the PG-rich cosolvent systems as noted by Herkenne et al. [18] and Manrique et al. [17]. The observed solubilities were lower than those previously reported for ethanol:water systems [14] since PG is a more polar solvent than ethanol [19, 20].

The amount of PG in the PG/water mixtures clearly influenced ibuprofen delivery into silicone membrane

and flux values were optimal for PG:water systems containing 70% PG. If PG enhances permeation by altering the solubility of the permeant in the membrane as proposed by Kasting et al. [11], alterations in the thermodynamic parameter would be anticipated; however, no distinct trends are apparent in the $K \times h$ values for binary systems in silicone membrane.

As for the permeation rate in silicone, as the amount of PG in the formulation increased, so did the permeation rate in skin for binary PG:water systems. The flux increase is particularly noticeable between 25 and 50% PG. Overall, flux values from PG:water are lower than from corresponding ethanol:water mixtures as reported previously [14, 20]. The derived kinetic parameters are independent of the vehicle composition. In contrast, the thermodynamic parameters derived for the formulations containing 90 and 25% PG are significantly different from those with 50 and 100% PG, and the values of k_p reflect these differences too. Overall, the results are consistent with recently published in vivo data [18], which showed that ibuprofen delivery into the stratum corneum increased with increasing percentage of PG in PG:water formulations and that the differences observed were because of changes in the partition coefficient, not the drug's diffusivity across the stratum corneum. In addition, a clear decrease in K with increasing amounts of PG in the vehicle was also observed by these workers.

The data from the non-occluded study with ternary vehicles and silicone suggests the formation of a supersaturated state for ibuprofen as ethanol evaporated from the vehicle. The flux from the system with lower ethanol content was significantly greater when there was no occlusion relative to occluded systems ($p < 0.05$), which suggests further that changes in the ethanol content of this system impacted on the thermodynamic state of the drug. The k_p value for the non-occluded system with lower ethanol is significantly higher than the corresponding k_p value for the occluded system, and although no dif-

ferences are seen in the k_p values for the systems with higher ethanol content, evaporation may have less of an impact on the ethanol content of these systems. These formulations have an ethanol content of 50% and consequently less influence on the drug permeation characteristics might be expected. Flux values for the formulations with the higher proportion of ethanol were statistically the same whether occluded or non-occluded, also suggesting that at higher concentrations evaporation does not play such a significant role. The ethanol content appears to be the critical determinant of ibuprofen flux enhancement for non-occluded systems in silicone.

Ternary systems with ethanol 50% v/v had a pronounced effect on ibuprofen permeation in skin, whereas no statistically significant differences were observed between the two ternary formulations when evaluated with silicone membrane. This may reflect the direct effects of ethanol on human skin as compared with silicone membrane. Ethanol is known to extract skin lipids [21], which should enhance solute flux. However, as we have noted previously, while flux enhancement is indeed observed

for ethanol-water mixtures, pure ethanol appears to retard ibuprofen permeation presumably because of its ability to dehydrate the skin [14, 22].

In conclusion, for simple PG:water systems, increasing amounts of PG enhanced ibuprofen permeation in silicone with optimal levels at 70% PG, but in skin, ibuprofen permeation systematically increased with increasing PG. An analysis of the kinetic and thermodynamic parameters for experiments conducted on skin confirmed that the influence of PG in the simple PG:water systems is primarily on the solubility and partitioning behaviour of ibuprofen. The corresponding analysis for the ternary ethanol:PG:water system confirms that the inclusion of ethanol will also modulate the partitioning and solubility behaviour of ibuprofen in skin.

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